

Portfolio

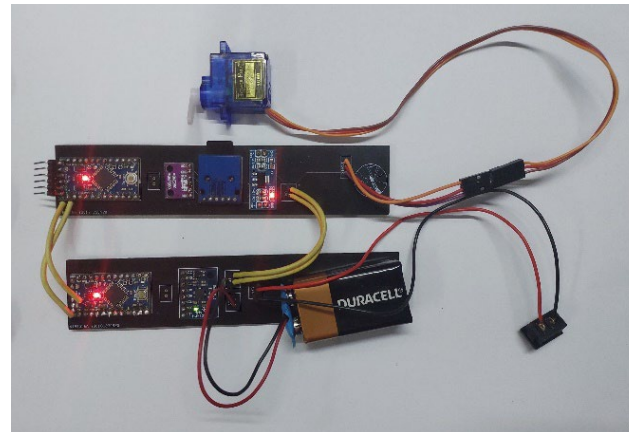
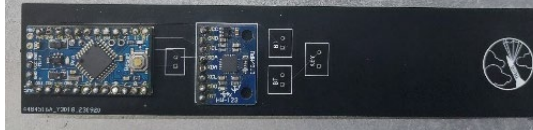
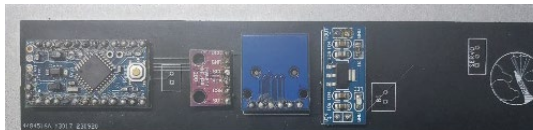
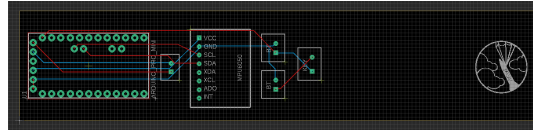
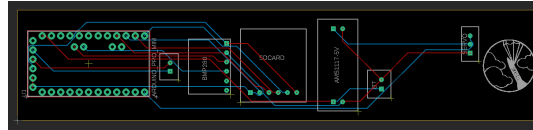
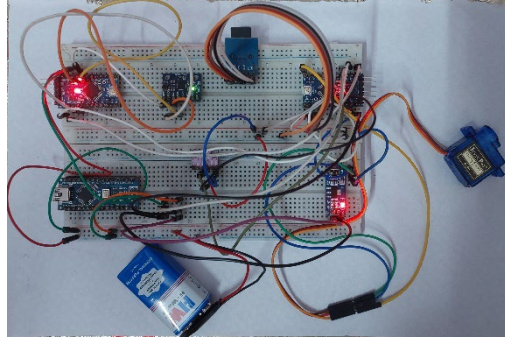
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ROCKET SKYFALL



Design and Fabrication of PCB



Manufacturing the Rocket and Solid Rocket Motor



Sounding rocket with an avionics bay as the telemetry data acquisition unit whose data was later analyzed using MATLAB App to plot the 3D trajectory of the rocket using the Inertial Measurement Unit (IMU) data

Rocket
Length 90 cm, max. diameter 5 cm
Mass with no motors 821 g
Mass with motors 1061 g

OpenRocket Simulation

Stability: 1.22 cal
CG: 59.4 cm
CP: 65.5 cm
at M=0.30



Apogee: 387 m
Max. velocity: 87.8 m/s (Mach 0.26)
Max. acceleration: 82.5 m/s²

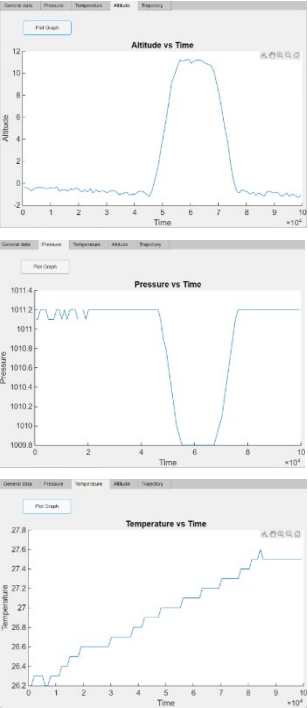
MATLAB App for Analysis

MATLAB App

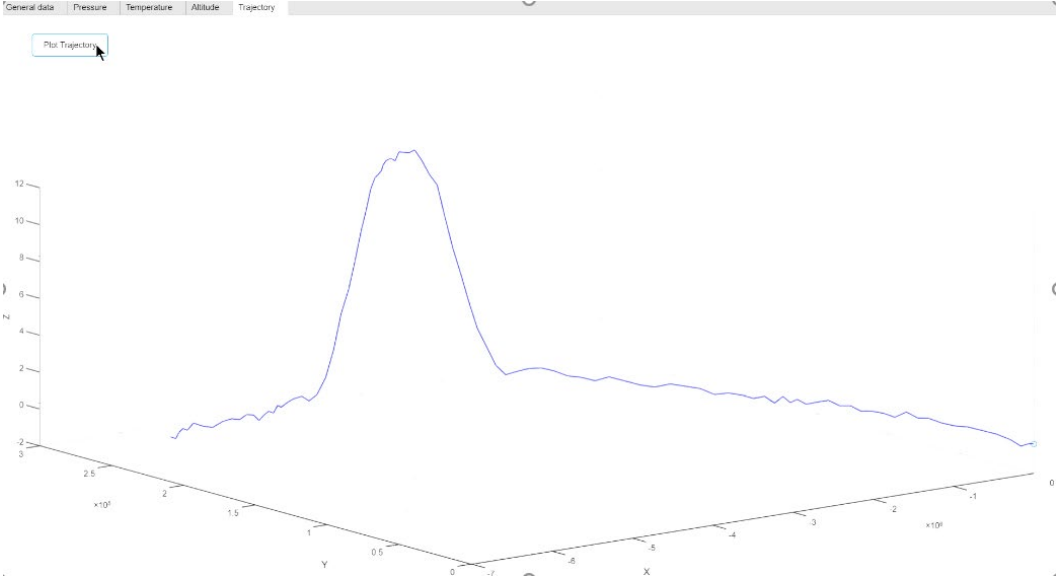
General data | Pressure | Temperature | Altitude | Trajectory

Import File

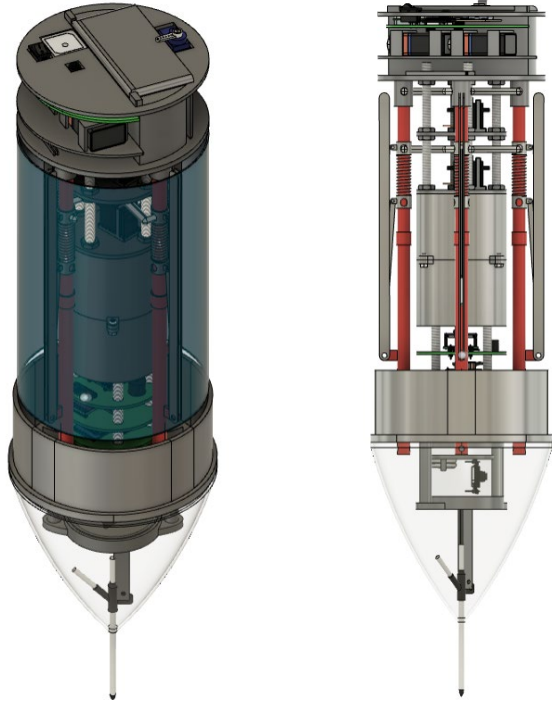
Time	Pressure	Temperature	Altitude	Acc x	Acc y	Acc z	Gyro x	Gyro y	Gyro z
47211	1.0111e+03	26.9000	0.3000	-608	-420	21412	277	575	1076
48217	1.0109e+03	27.0000	1.3000	-1644	-64	20580	-487	-369	1048
49217	1.0108e+03	27.0000	2.8000	-956	-428	20308	182	370	-28
50225	1.0106e+03	27.0000	4.3000	-696	-620	20332	-666	-481	-1284
51220	1.0104e+03	27.0000	5.7000	-36	-56	20308	63	93	-209
52228	1.0102e+03	27.0000	7.4000	-676	-28	20048	-537	-98	-291
53230	1010	27.0000	9.2000	-672	16	20064	-142	-147	146
54236	1.0099e+03	27.0000	9.8000	-564	-60	19248	-228	-490	-137
55237	1.0098e+03	27.0000	10.6000	-520	-212	19252	-359	55	218
56243	1.0098e+03	27.1000	11.2000	-40	-408	20500	-648	-239	-545
57246	1.0098e+03	27.1000	11.1000	152	-788	20676	-360	-478	-778
58249	1.0098e+03	27.1000	11.2000	-44	-496	20212	-96	-373	-128
59249	1.0098e+03	27.1000	11.3000	164	-328	20076	246	49	-85
60257	1.0098e+03	27.1000	10.9000	-236	-600	20184	-274	-331	466



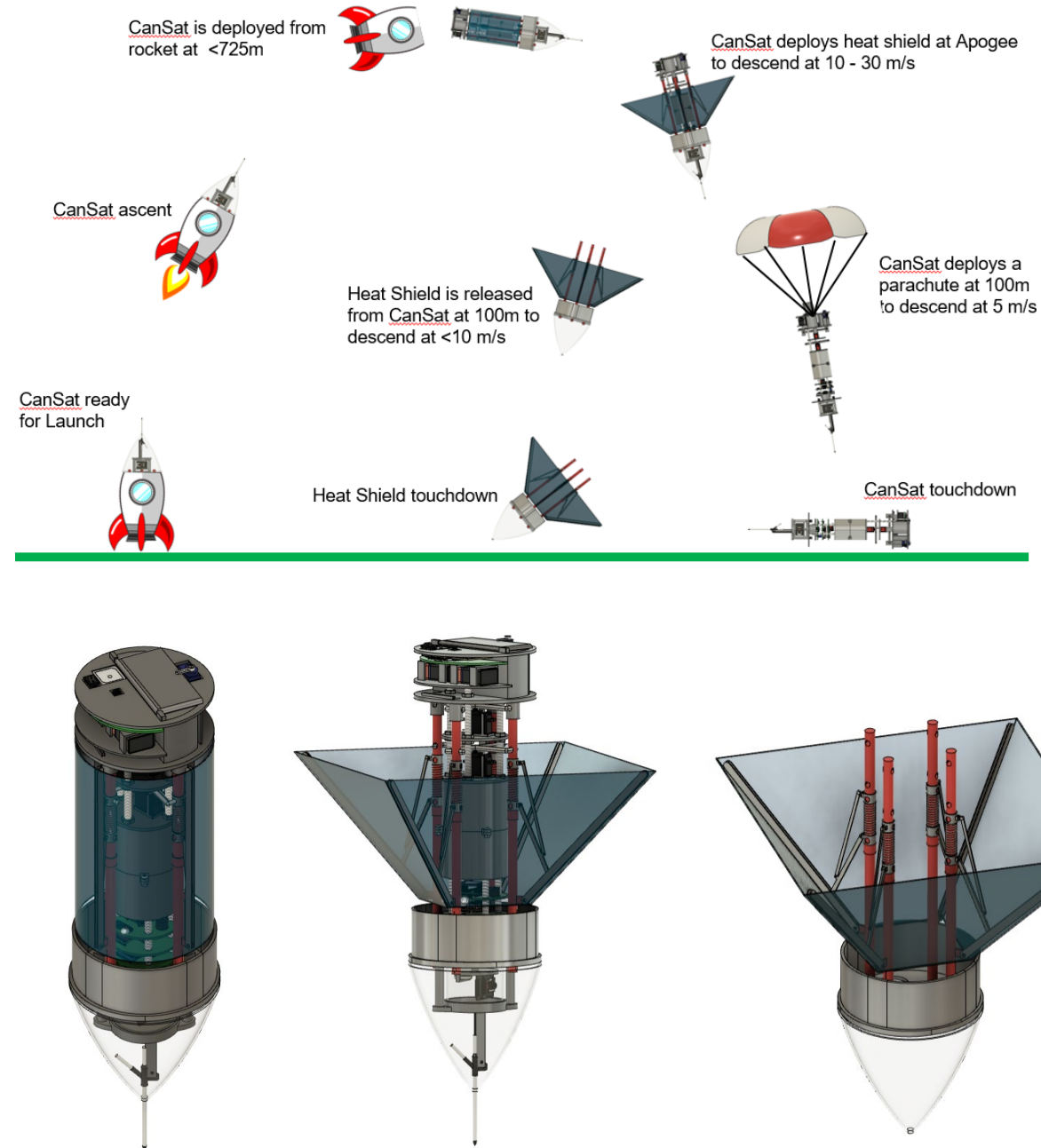
3D Trajectory plot



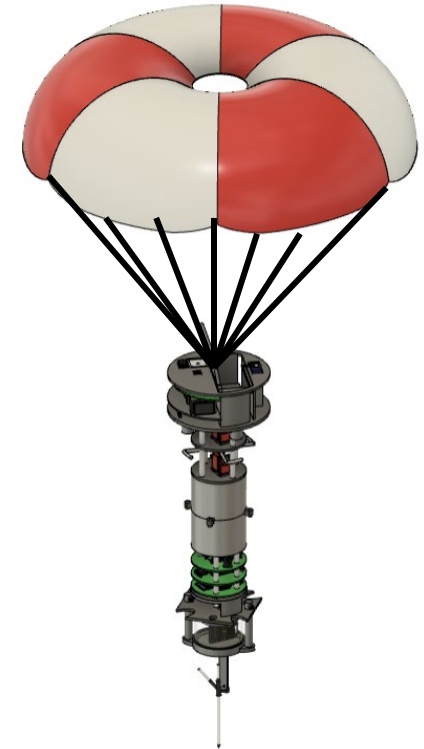
CanSat 2024 Team CANOPY



The objective of the competition was to design a CanSat that ejects a heat shield to break the fall and subsequently deploys a parachute to safely land an egg that simulates a delicate object safely from an altitude of 725m



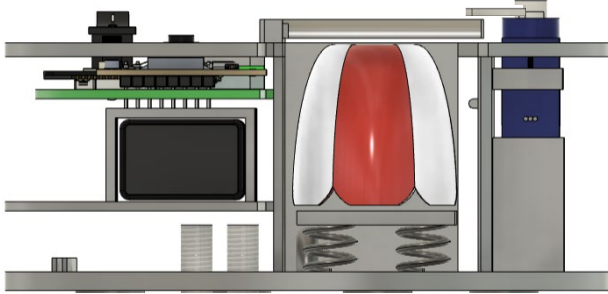
Descent Control for the Mission



Parachute Deployment Mechanism

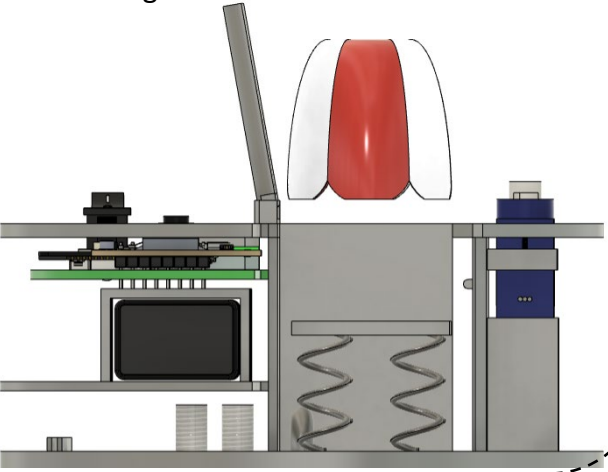
Parachute kept under tension by spring plate at bottom

STOWED CONFIG

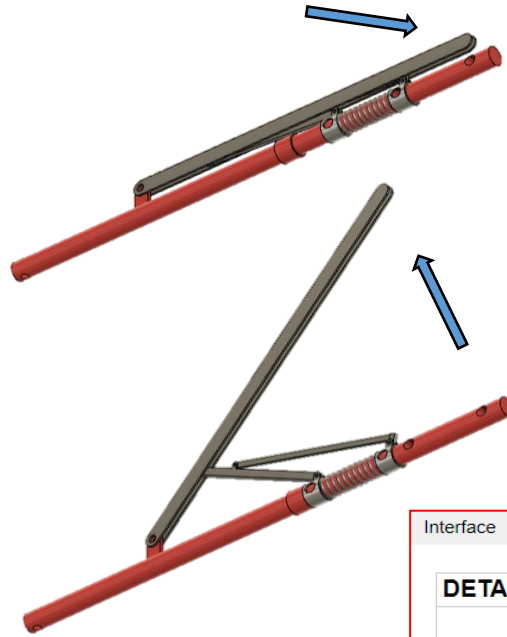


Springs spring open pushing the parachute out of which is hinged inside at the centre

DEPLOYED CONFIG



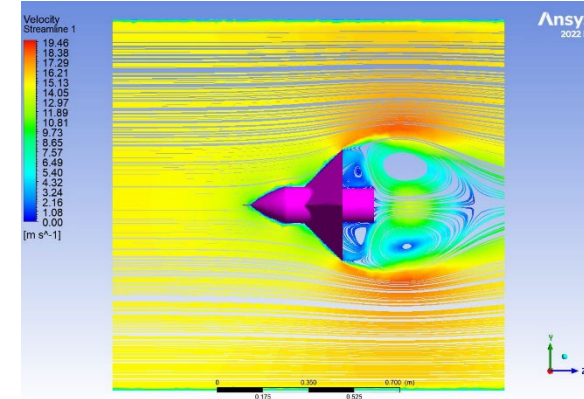
Heat Shield Deployment



Egg Protection Chamber



Velocity Profile of CanSat



Avionics Bay



MATLAB UI for Ground Station

Interface

CANSAT data

Plotted Data 1

Plotted Data 2

GPS

+

DETAILS

Mission Date

mm/dd/yyyy

Mission Time (UTC)

0

Team No.

2071

Battery %

0

Packet Count:

0

☒ Connection Status

Enter Command

Send Command

MISSION TOOLBOX

Generate Mission Report

Generate .csv file

Flight ☐ Simulation

Mode Indicator

MISSION STATUS

Pre-Launch

Launch

Apogee

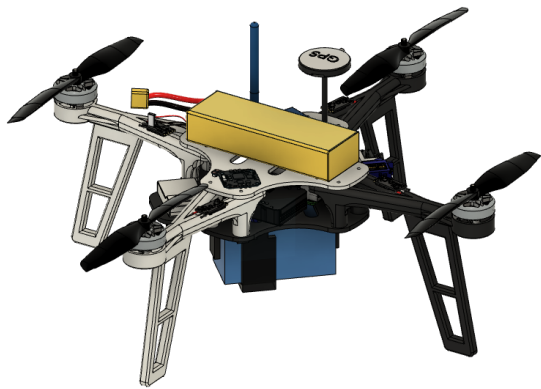
Heat Shield deployed

Heat Shield released

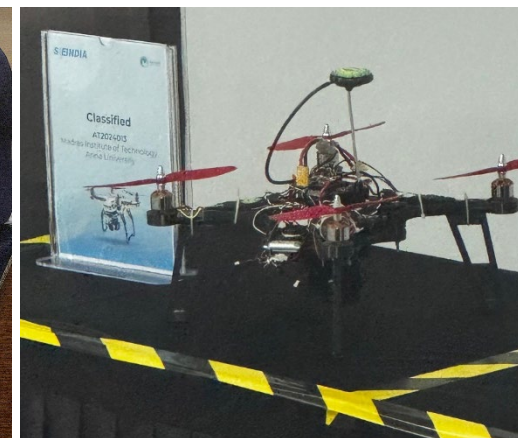
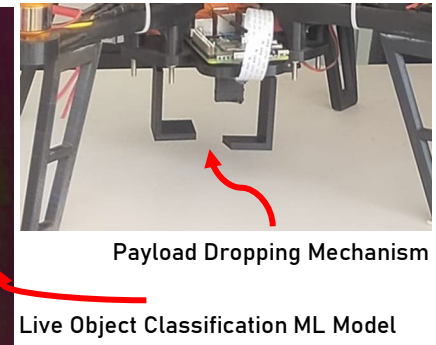
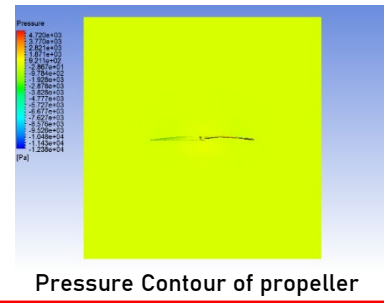
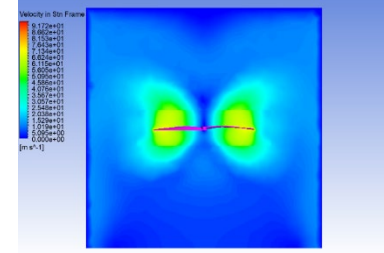
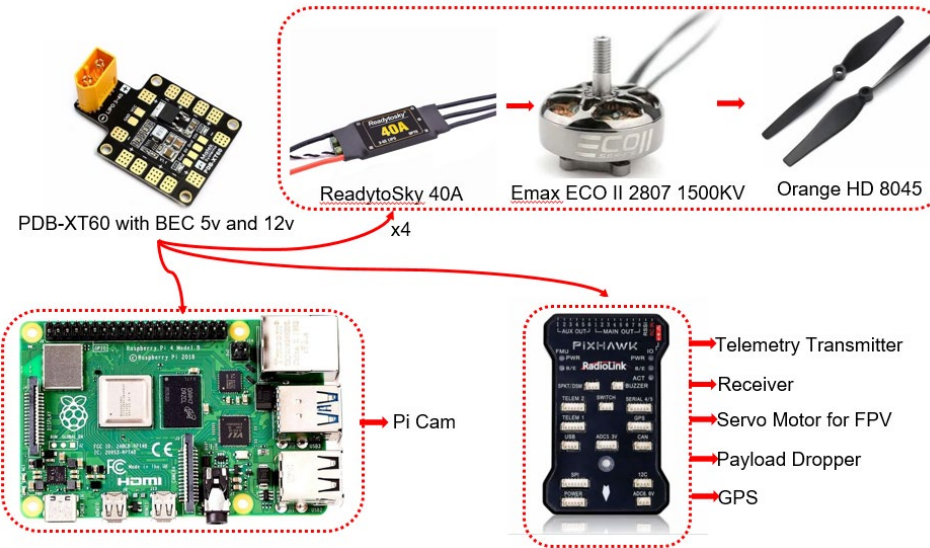
Parachute deployed

Landed

AeroTHON 2024 Team Classified



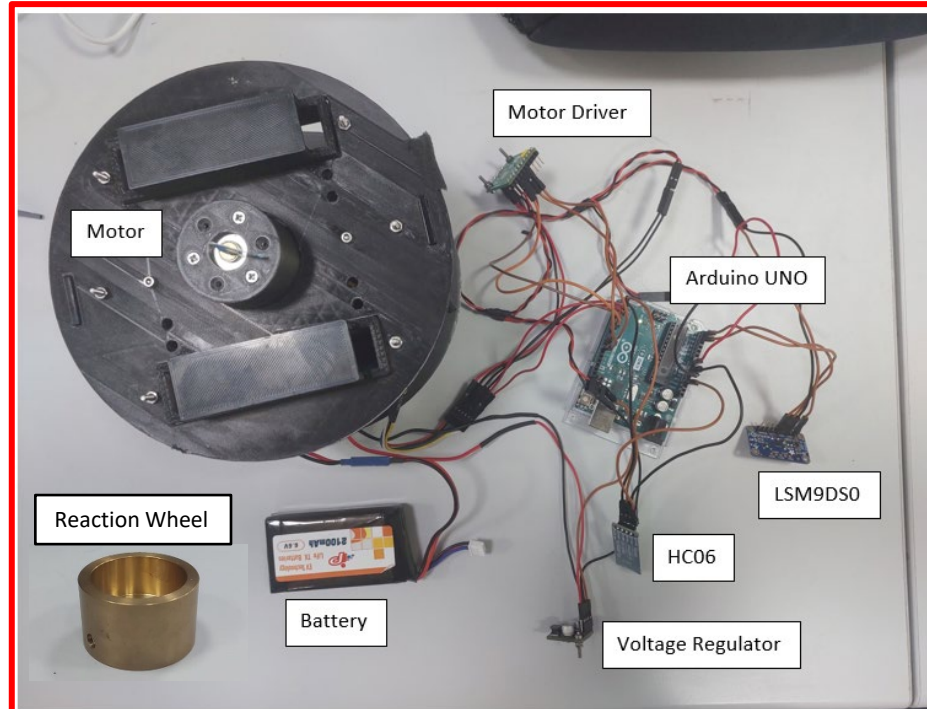
Designed an autonomous drone with payload dropping and terrain mapping and object classification capabilities to safely drop a payload at a designated spot.





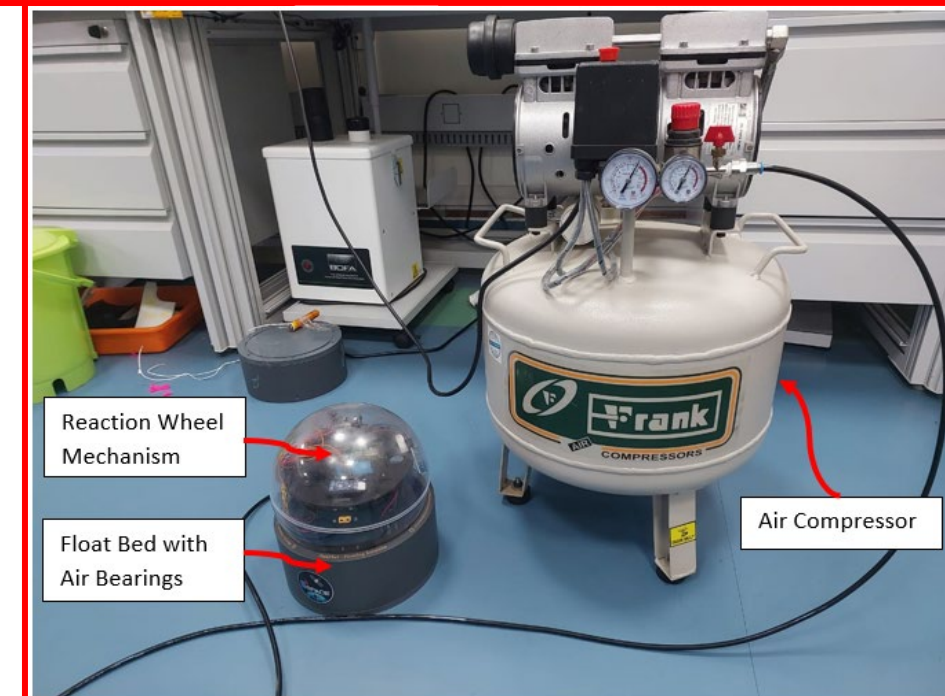
Satellite Attitude Determination and Control System

Developed and implemented the control algorithm of one DOF for a single reaction wheel on a FloatSat with air bearings to test the Attitude Determination and Control Subsystem on CubeSats to achieve precise Z-axis stabilization and control



Components Overview

Experimental Setup



FloatSat and Air Bearing



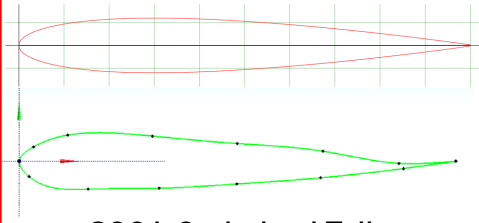
Anna University

Madras Institute of Technology (MIT) Campus

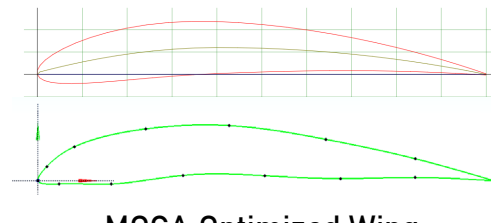
Methods of Stabilization for Ground Effect Vehicles

Engineered stabilization methods for Wing-In-Ground Effect Aircrafts to improve their longitudinal stability via

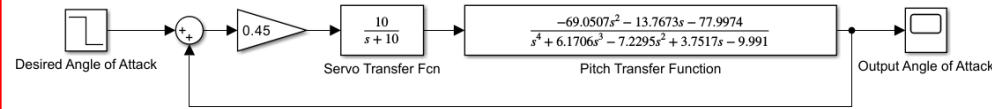
1. CFD based on SOGA Optimization (Single-Objective Genetic Algorithm) of horizontal stabilizer.
2. Feedback Control System for Pitch Damping in the form of a pitch damper.



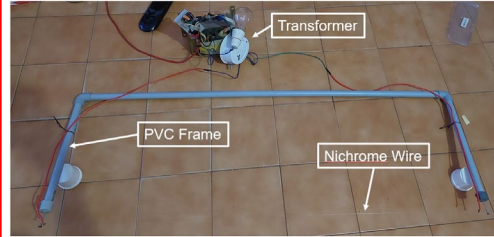
SOGA Optimized Tail



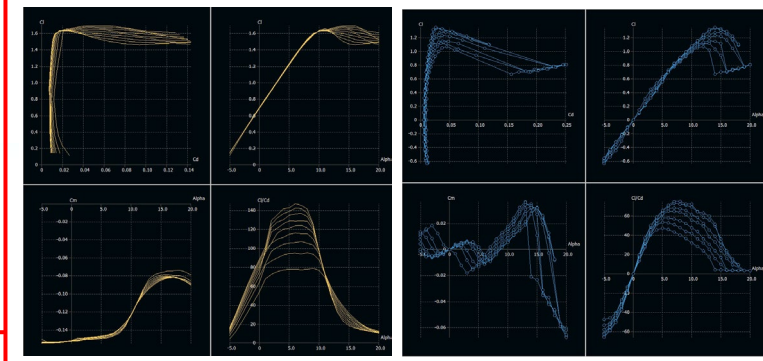
MOGA Optimized Wing



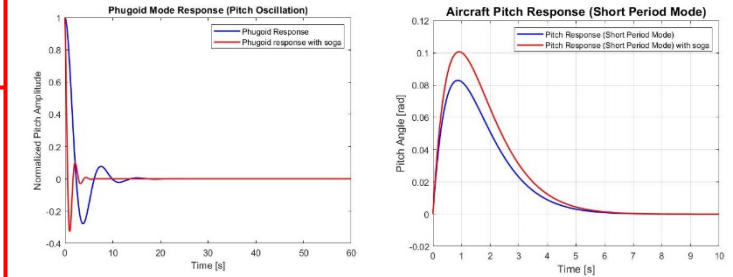
Pitch Damper from Transfer Function



BUILD



XFLR5 Analyses



Stability Modes Analyses



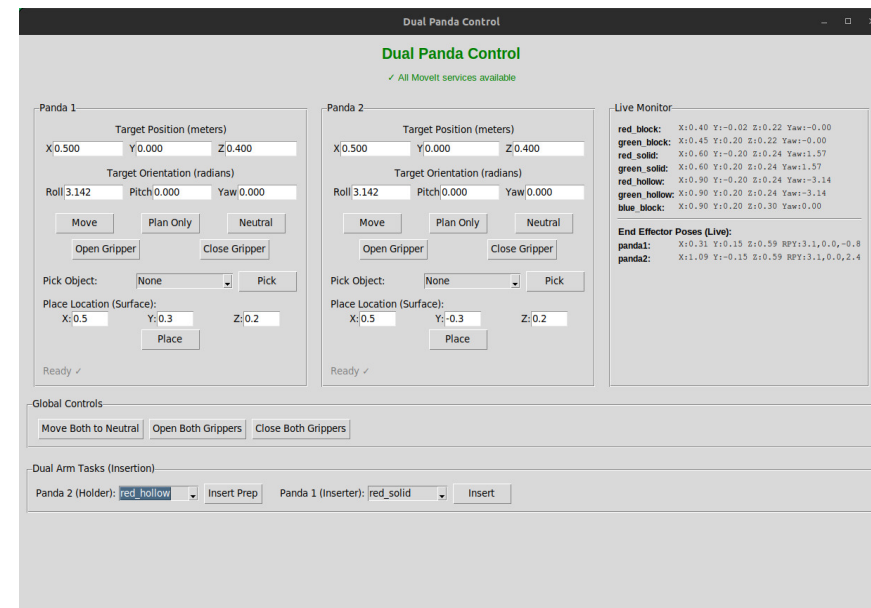
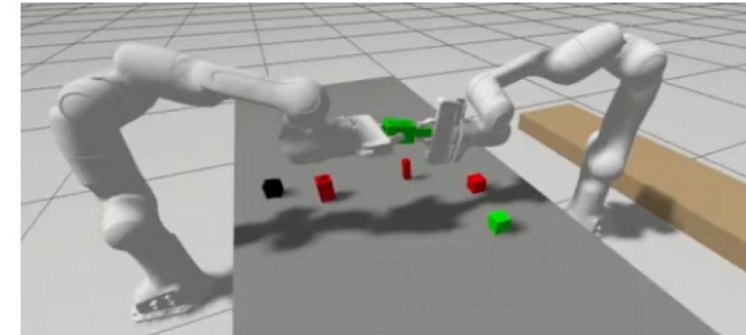
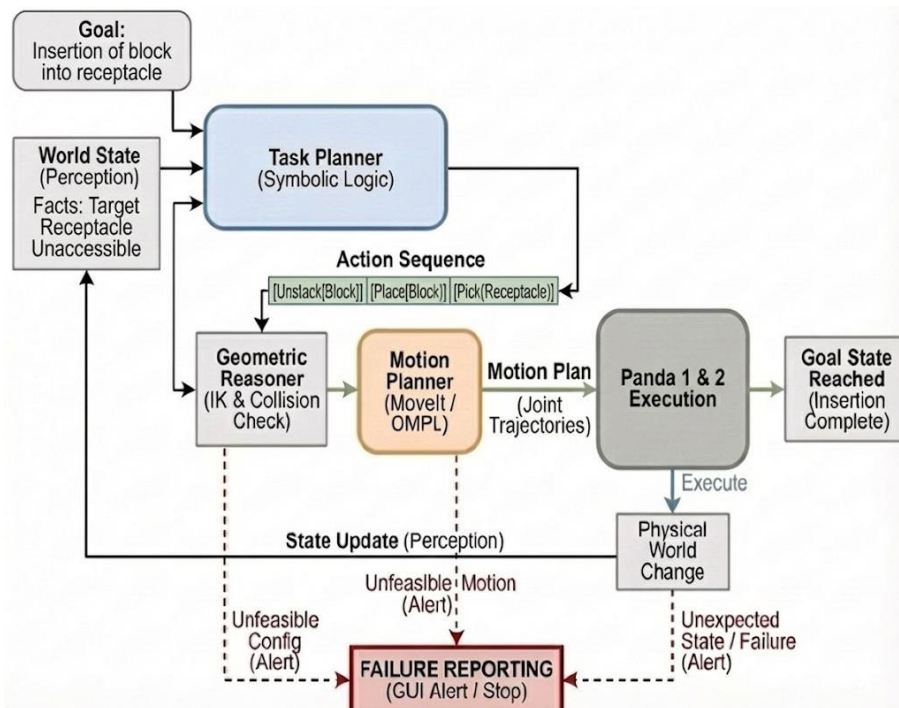
Ground Effect Testing



Team CANTFLY

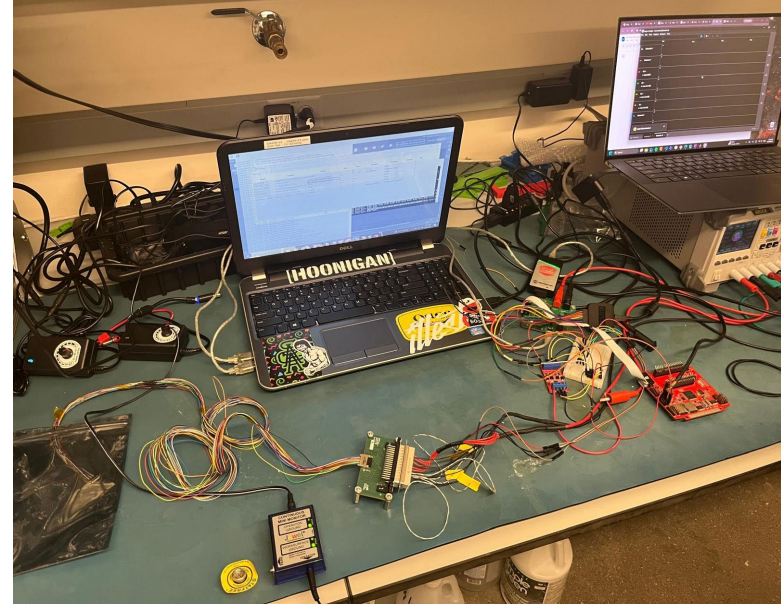


Performed a coordinated insertion task of a peg-in-hole using two franka manipulators using Task and Motion Planning (TAMP) via SITL



SWARM-EX CubeSat

Developing GCS software for the ADCS subsystem of the SWARM-EX CubeSat involving the ADCS component XACT-15 of Blue Canyon Technology.

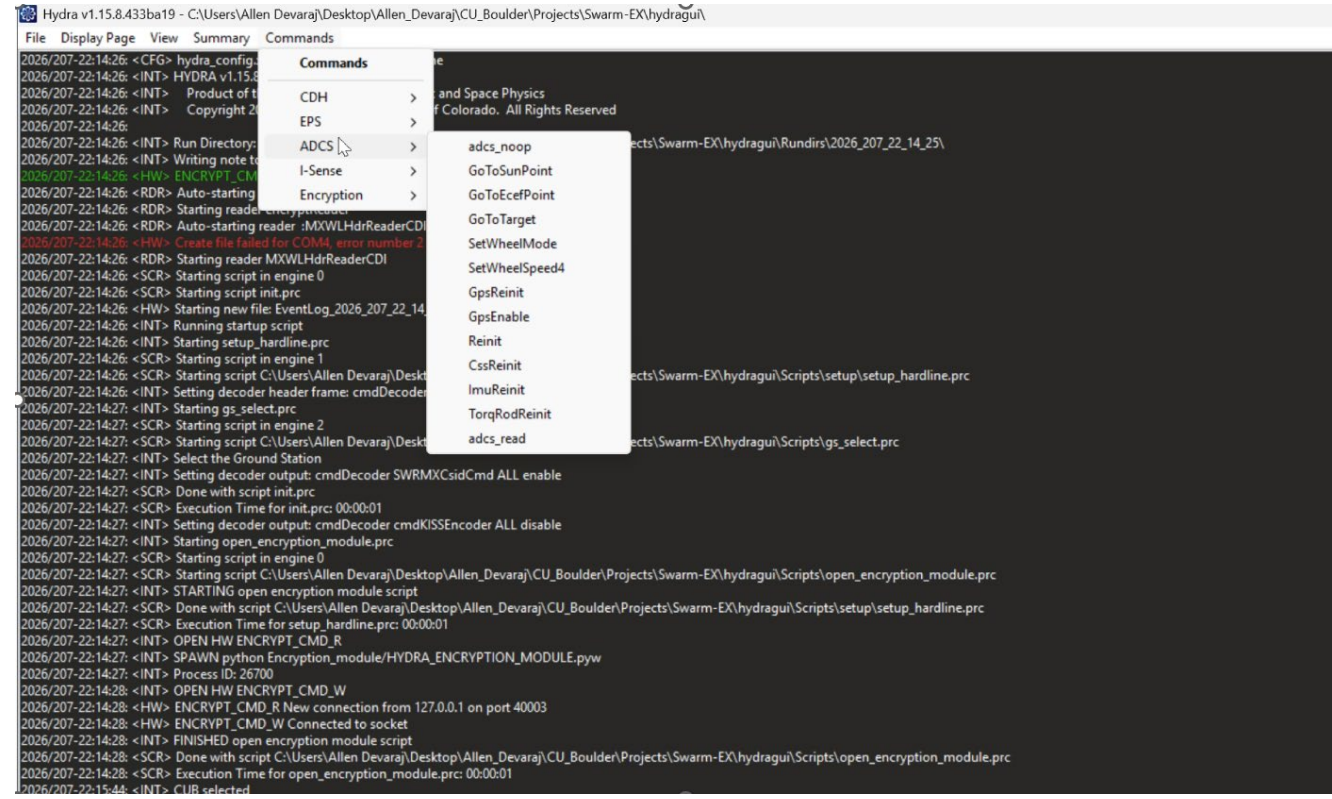


FlatSat setup

XACT-15



Hydra





University of Colorado **Boulder**



Correll lab

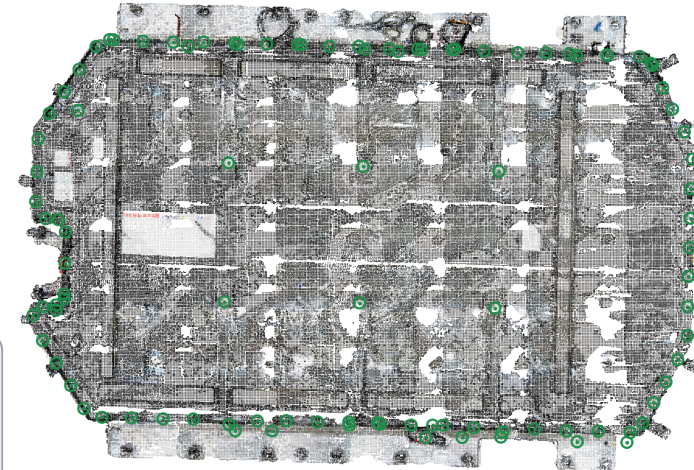
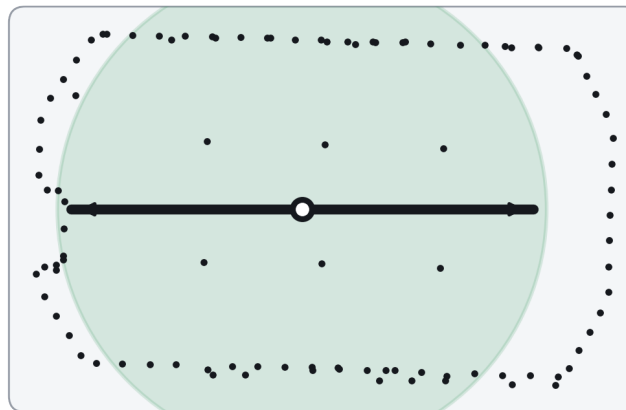
COLLEGE OF ENGINEERING AND APPLIED SCIENCE

Robotic Disassembly of EV Car Battery

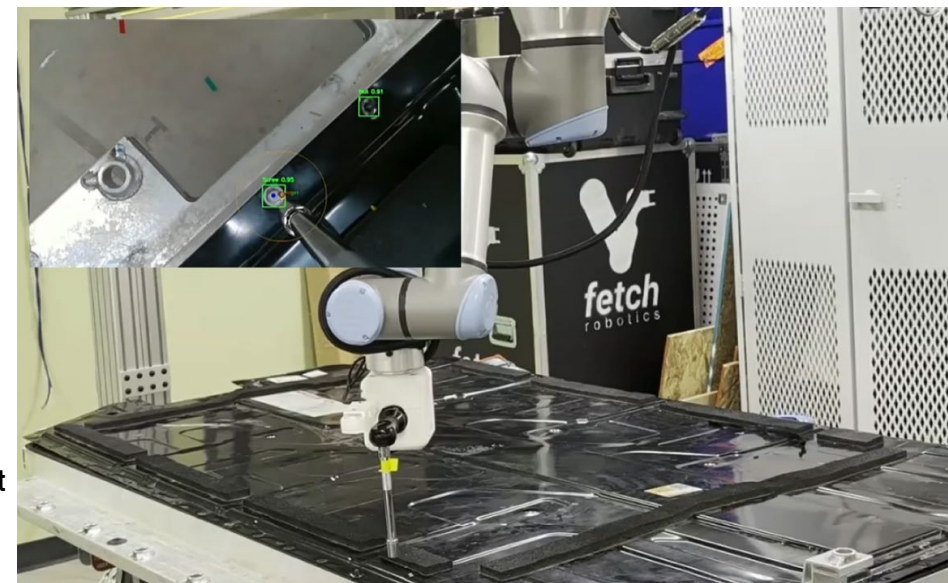
Developed a constrained motion planning pipeline with analytical IK seed ranking and cost-normalized configuration selection for a 7-DOF industrial manipulator system.



Screw distribution with respect to the gantry and joint space extension using the linear actuator



Pointcloud fed to the VLM



VLM reasons about the screws left to be unscrewed



University of Colorado **Boulder**

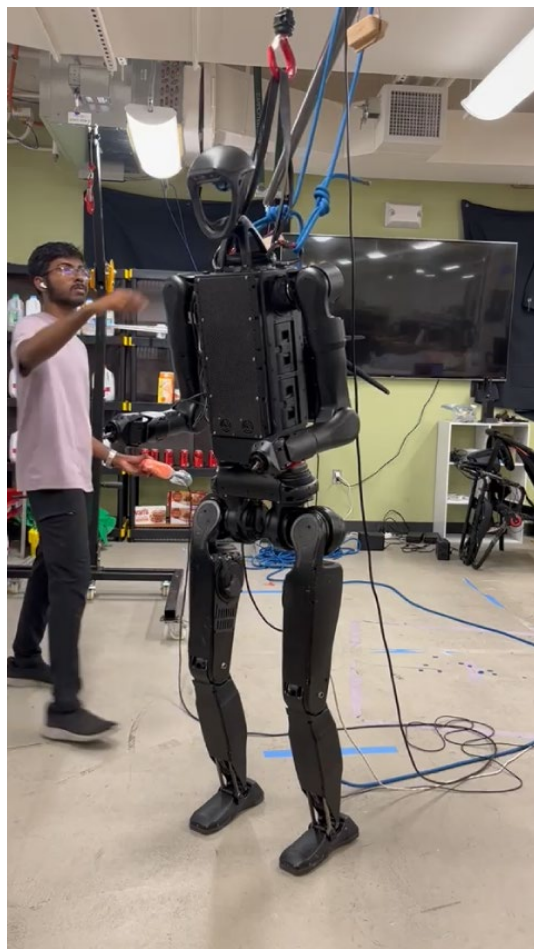


Correll lab

COLLEGE OF ENGINEERING AND APPLIED SCIENCE

Humanoid Locomotion with MPC

Deployed MPC based sampling on a Unitree H1-2 for locomotion.



Contact Implicit MPC



working
towards
Loco-Manipulation

Push
Recovery
Testing

